

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CE)

November – December 2011

CE 202 Data Communication and Networking

Date: 30.11.2011, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Answer the following questions.

[04]

1) Telephone systems may be classified as:

- A) Simplex and asymmetrical B) duplex and symmetrical.
C) Duplex and asymmetrical. D) Simplex and symmetrical.

2) A router:

- A) Forwards a packet to the next free outgoing link.
B) Determines on which outgoing link a packet is to be forwarded.
C) Forwards a packet to all outgoing links.
D) Forwards a packet to all outgoing links, except the incoming.

3) The Internet is an example of a:

- A) Circuit switched network. C) Packet switched network.
B) Cell switched network. D) None of the above.

4) A distributed network configuration in which all data/information pass through a central computer is:

- A) Bus Network B) RING Network C) Star Network D) None of Above

(b) Explain Data Communication Components with examples.

[03]

Q - 2 (a) List Different types of Addressing mechanisms with LAYER those are used in computer network.

[03]

(b) Explain Hamming code with following data. Sender's data = 11101011 and Receiver's data = 11001011

[06]

(c) Compare MESH topology with STAR topology.

[05]

OR

Q - 2 (a) What do you mean by Flow control? Explain Stop-n-wait ARQ?

[06]

(b) What do you mean by error control? Explain CRC with an example?

[06]

(c) Why SWITCH is better than HUB?

[02]

- Q - 3 (a) Explain Guided transmission mediums. [06]
- (b) 1. Explain different Types of Connection. [06]
2. Explain simplex, half-duplex, full-duplex with an example.
- (c) Suppose the IP address of the system is 172.16.24.162/20 what is the broadcast address of this network. [02]

OR

- Q - 3 (a) Write down brief notes on duties of PRESENTATION LAYER. [06]
- (b) Short note on CSMA/CA. [06]
- (c) List any four protocols those works at Application Layer. [02]

SECTION – II

- Q - 4 (a) Explain Null MODEM with its advantages [04]
(b) Compare Analog signal with Digital signal. [03]
- Q - 5 (a) Explain asynchronous RS-232 in detail. [07]
(b) Which Modulation techniques are generally used in MODEM? Why? [03]
(c) Assume the data stream: 10100101010010100 represent this data stream using 1) Manchester 2) Differential Manchester. [04]

OR

- Q - 5 (a) Explain various techniques of serial data transmission. [07]
(b) Define 1) Bandwidth 2) Phase 3) Baud rate. [03]
(c) Describe the characteristics of sine wave. [04]
- Q - 6 (a) Differentiates LAN, MAN, WAN. [04]
(b) An organization has a class B network and wishes to form subnets for 20 departments. What is the subnet mask? [02]
(c) Why the size of window is $2^n - 1$ instead of 2^n in Go-Back-N protocol? Justify. [02]
(d) Explain cable modem in details. [04]
(e) List the application of Multiplexing. [02]

OR

- Q - 6 (a) Define 1) Routing 2) Aperiodic analog signal [04]
(b) List the application in which it requires to perform analog to analog conversion and list the techniques of performing analog to analog conversion. [02]
(c) Why the size of window is $2^n/2$ instead of $2^n - 1$ in Selective Repeat protocol? Justify. [02]
(d) Compare DTE with DCE. [03]
(e) Explain Checksum method with an example. [03]
